Installation Manual

Trimble® Earthworks Soil Compactor

For SITECH Only

Version 2.24.x
Revision A
March 2026

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Compactor installation

In this guide:

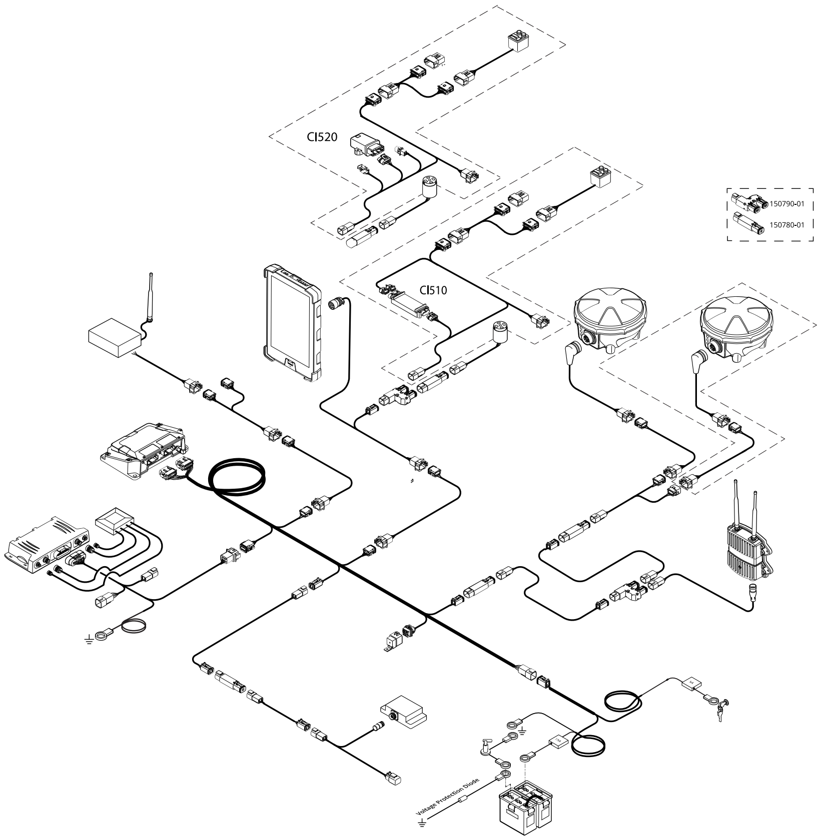
- Installation introduction
- System components, cables and connections
- Kit list
- Welding and fitting tasks

1.1 Installation introduction

This manual describes how to install the Trimble® Earthworks system on a soil compactor.

1.2 System components, cables and connections

1.2.1 Generic install



1.3 Kit list

Use the following kits to install the compactor components:

Kit number	Description
160075-500	Base, Generic, Compactor Platform, V3.0
200435-10	EC520 Control Module
58440-10	Mount – RAM Ratchet Clamp + Ball
597-1269, or 110262-00	Display Mounting Post, or Bracket – Display, Magnetic Mount
74708-09	Install, Compactor, Trimble Ready, CM310 (Optional)
64861	Plate Sensor Mounting CM310 (only required for CM310)
121022	Kit – Voltage Protection (only required for CM310)
160041-10	Kit – SNM941 Mounting, Roofline Tandem Mounting (optional)
112057-00	Kit – SNM941 4-in-1 Antenna, 0.6m & Mounting Bracket (optional)
–	GNSS receiver or receivers: MS952, MS955, MS972, MS975, MS992, or MS995 (optional)
59701-62	MT9xx UTS target (optional)
–	TD5x0 display. You may swap the TD5x0 for a third party tablet. For information on supported BYOD tablets, refer to the *Trimble Earthworks Release Notes*.

1.4 Welding and fitting tasks

Possible welding and fitting tasks are outlined in:

- *Compactor Cab Install Guide*
- *Compactor Additional Sensor Install Guide*
- *Compactor GNSS Install Guide*

1.4.1 Hints and tips for welding

WARNING – Disconnect the negative battery terminal before commencing any welding and ensure that power is removed from the grade control system by disconnecting the power input cable or cables. Failure to follow this warning may result in electrical damage to system components.

TIP –Tack weld and then check clearance and proper alignment before welding completely.

Decide where to locate each bracket, then arrange for someone to weld the brackets in place. Welding minimizes the amount of vibration and keeps the component's pattern of vibration similar to the pattern of vibration of the machine.

Consider the following:

- Some brackets have small "window" cutouts. Use a wire feed welder to reach into these areas.
- After welding, paint or repaint the welded surfaces to control corrosion.

2

Compactor installation checklists

In this guide:

2.1 Checklist—Pre-installation

Use the following checks to identify problems with a machine before installation.

- ☐ If the machine is manufactured by Caterpillar, check whether it has a factory-installed CMV or MDP sensor connected to the machine ECM. This equipment is required for compaction mapping. See 9 Compactor machine ECM install guide for more information.
- ☐ If the machine is manufactured by Dynapac, Bomag, or Hamm, check whether the optional Dyn@Lyzer machine ECM, Bomag machine ECM, or Hamm machine ECM and its firmware and the mandatory required hardware are installed. These equipment are required in order to connect to Earthworks.
- ☐ Contact the machine manufacturer to ensure that the machine ECM firmware is up to date.
- ☐ If you intend to install the compaction sensor, ensure that there is space to fit the bracket to the side of the drum:
  - Position the bracket on the hub of the machine drum.
  - You must install the bracket on the:
    - inner side of the drum shock mounts
    - side of the drum that does not rotate
  - Do not isolate the compaction sensor from the drum vibration.

3

Tools and other supplies

Make sure that you have the following tools and supplies available:

- Arc welder / welding rods
- Volt/Ohm meter
- Mechanical and electrical hand tools
- Clean container for hydraulic oil
- Clean rags

4

Compactor cab install guide

In this guide:

4.1 Overview

This guide describes how to install the components in and around the cab on a compactor system.

> **TIP –** There are many tasks, such as installing or updating firmware and licenses, that need to be performed for an installation. Often these tasks are most efficiently performed in the office before going out to site.
>
> It is strongly recommended that you read this manual carefully to familiarize yourself with the installation procedure, and to identify the tasks that can be done before going out to site.
>
> Some tasks, such as updating the EC520 firmware, may require simple cables to be built in order to connect the device to an external power supply. Alternatively, assemble the system harness to the extent required to power these devices.

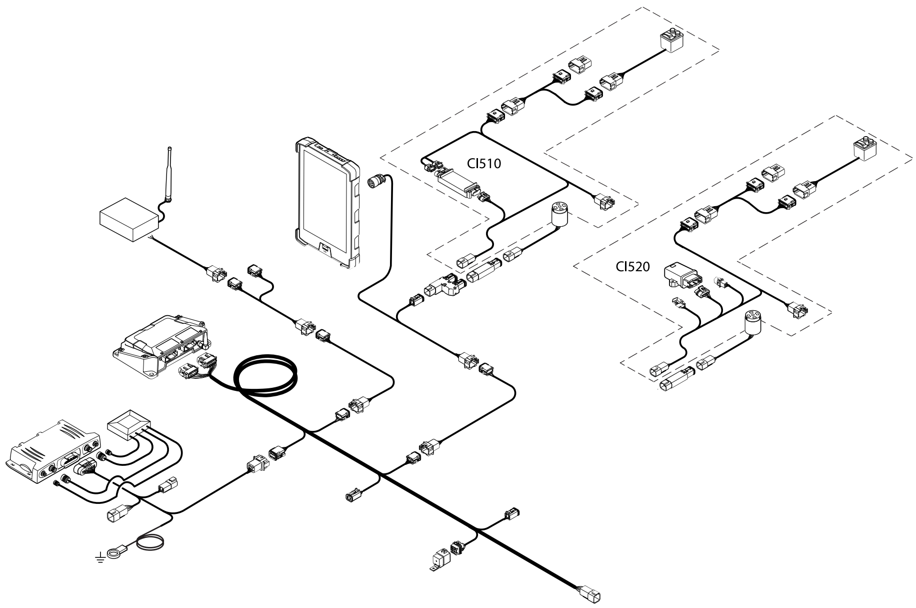
4.1.1 Cable routing best practice

When selecting a route for cables and harnessing consider the following:

- Exposure to hazards, including:
  - mechanical damage by the operator
  - environmental hazards
  - heat
  - corrosive material

- Moving parts
- Sharp edges
- Access to the AA510 audible alarm

4.1.2 Cab components, cables and connections

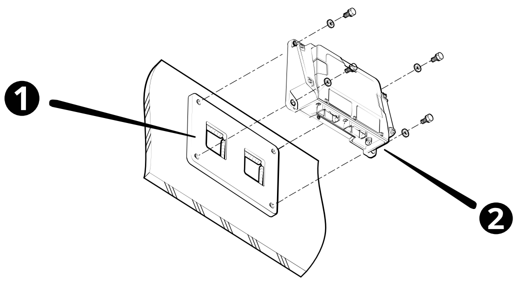


4.1.3 Prerequisites

- The kit includes a 12V relay (P/N 109333-12) and a 24V relay (P/N 109333-24). Only use the relay that matches your machine's voltage.
- Do not remove the backing film or the sorbothane layer on the EC520. The film protects the sorbothane layer.
- Do not cut the cables. Use extension cables where needed.

4.2 EC520 control module install

This section describes how to install the EC520 control module.



Item	Part number	Description
❶	142220-30	EC520 bracket
❷	200435-10	EC520 control module

4.2.1 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable

4.2.2 Install the EC520 control module

Requirements:

- Position the EC520 bracket to a rigid section of the Engine End Frame (EEF), avoiding thin sheet metal, poorly supported areas, or the isolated cab. The EC520 must vibrate with the machine body.
- The EC520 measures the pitch and roll of the machine. To ensure this is accurate, install the EC520 either:
  - Parallel to the centerline of the machine
  - Perpendicular to the centerline of the machine
- The connectors may be facing left, right, forward, back, or down. Avoid mounting the EC520 on an inclined surface, or with the connectors facing upwards.
- If the EC520 is moved or replaced at any time, check the Drum Northing, Easting and Elevation, Body Mainfall, and Body Slope values under Machine Diagnostics. If the values are not accurate, perform the Measure-up section of the Install Assistant again to re-calibrate the pitch and roll.

Suggested locations for the EC520:

- Behind the cab steps
- At the base of the cab body, toward the pivot point

To avoid damage:

- Do not install the EC520 with the ports facing up.
- Consider a location or surface that does not transfer heat to the EC520 or the harness.
- Allow space to route cables from the EC520 into the cab.

Steps:

1. Weld the bracket on the installation surface. Weld completely around the inside of the two window cutouts to secure the bracket. Do not weld around the outside

perimeter of the bracket.

You cannot drill into, or weld on, any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

> **NOTE –**Many cabs are now an integral part of the machine's ROPS, and are certified as such. Drilling into, or welding on, these cabs may invalidate their ROPS certification.

2. Attach the EC520 to the bracket using the screws and washers provided in the kit.

   > **NOTE –**Do not remove the backing film or the sorbothane layer on the EC520. The film protects the sorbothane layer.

3. Plug the two EC520 connectors (labeled EC520) into the module as shown in the connection diagram. See 4.1.2 Cab components, cables and connections.

   > **TIP –** The kit includes backshells and corrugated tubing for the EC520 Deutsch connectors. Use these to protect the connectors from water. When the backshell and tubing is installed, seal the end of the tubing with tape.

4.3 TD520 display install

This section describes how to install the TD520 display. There are 2 installation options:

- With the magnetic display mount (P/N 110262-00) and Ram ratchet ball-clamp, for cab window installations.
- With the post mount (P/N 597-1269) and Ram ratchet ball-clamp, for mounting on a vertical pole, such as a grab bar.

> **WARNING –**If you have a pacemaker and handle the magnetic display mount, the magnet can affect the function of the pacemaker and cause physical harm. Remain at least 30 cm (1 foot) from the magnetic display mount if you have a pacemaker.

> **WARNING –**The magnets on the magnetic display mount are strong. If parts of your hands get between two magnets when you move them, the magnets may suddenly move and pinch or clamp the skin. When the magnets are close together, position your hands so they are not between the two magnets.

If you are installing a TD540 display:

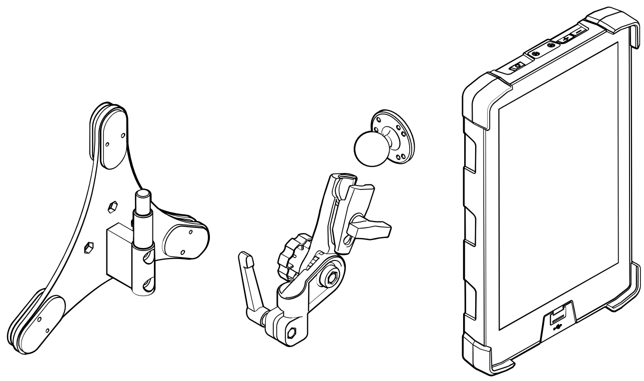
- Follow the steps below to mount the display in the cab.
- See the TD540 section for cable connection information.

> **NOTE –**It is easier to perform this task with two people.

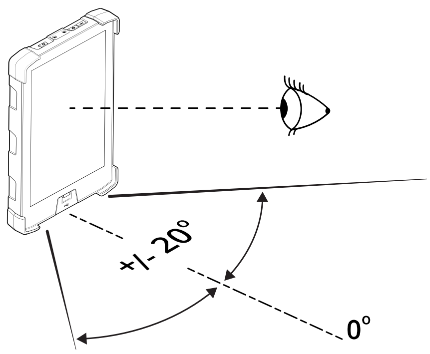
To select a route for the display cable, try various positions while considering:

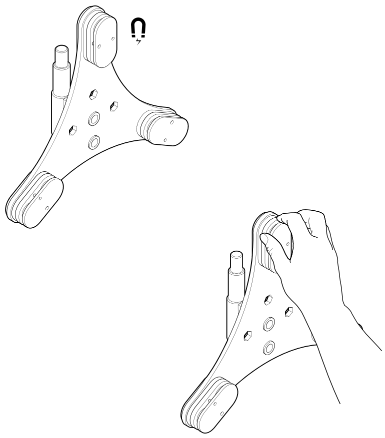
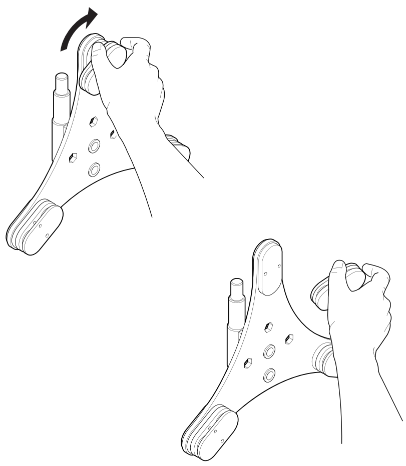
- Exposure to hazards, including:
  - mechanical damage by the operator
  - environmental hazards
  - heat
  - corrosive material
  - weather
- Existing grommeted entry points. When taking the cable outside the cab, use grommeted holes to minimize dust and moisture ingress to the cab.
- Maintaining free space between structures holding the display, or the display itself. For example, maintain free space between the display and windows that open and close, controls, and the view of the work area.
- Access to the AA510 audible alarm
- Avoid moving parts
- Avoid sharp edges
- Operator's preference for location and orientation

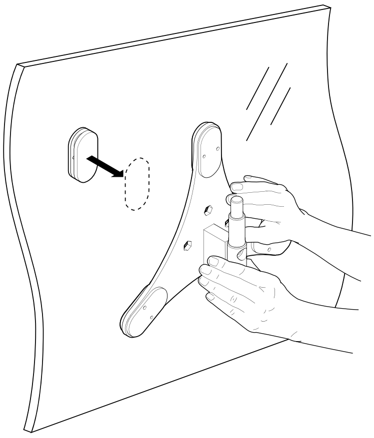
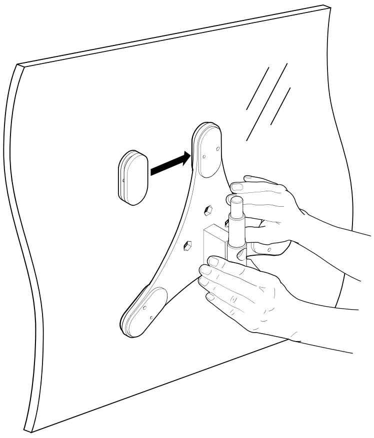
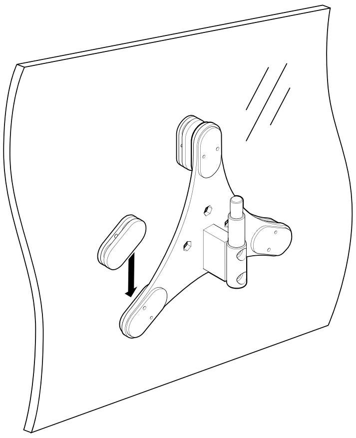
4.3.1 Option 1: Magnetic mount and display



Attach the magnetic mount

Steps	Description
	Decide on the location of the display.

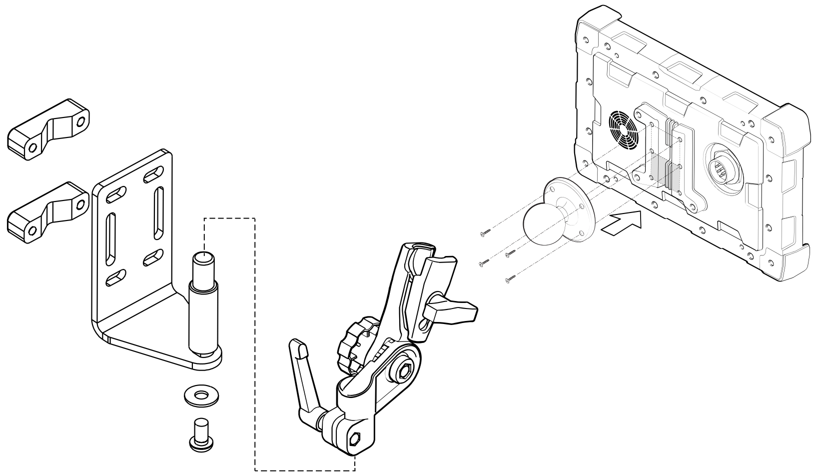
Steps	Description
	Hold the magnetic display mount and carefully twist the magnet and tri-arm in opposite directions.
	When the force between the magnets weakens, pull the tri-arm and magnet apart. Remove the padding and keep the magnets away from each other.
Practice connecting and disconnecting.	Using two people and the padding, practice placing the tri-arm and magnet together. Keep the magnets close but not opposite each other with the padding in between, then rotate and slide the magnet until the magnets connect.

Steps	Description
	On the machine, place the tri-arm magnetic assembly against the inside cab window in the required position. Check the post is aligned correctly and that the cables, Ram ratchet ball-clamp, and display are in the correct position. Remove the Ram ratchet ball-clamp and display before continuing. Hold the interior tri-arm firmly in the required position and make sure all three magnets touch the glass.
	On the outside of the glass, attach the first magnet. • Once the magnets connect they are very hard to disconnect. • Keep the magnet surfaces away from dirt or other metals. • Start with the outside magnet away from the inside magnet, then slide toward each other.
	Repeat for all three magnets.

To finish:

1. Attach the Ram ratchet ball-clamp.
2. Attach the display.
3. Attach the display cable and any extension cables required to reach the EC520 long platform harness, P/N 150603-02.

4.3.2 Option 2: Post mount and display



Attach the post mount

1. Connect the ball mount to the TD5x0.
2. Select a vertical post to mount the display to. The location should position the display within easy reach of the operator.
3. Use the appropriate M8 bolts and the V-blocks to attach the bracket to the post.
4. Fasten the RAM mount to the pole bracket.
5. Insert the ball mount into the RAM mount socket and fasten.
6. Attach the display cable and any extension cables required to reach the EC520 long platform harness, P/N 150603-02.

4.3.3 Validate

1. Turn on the ignition. The display starts automatically.
2. Tap the Earthworks icon. Earthworks opens.
3. If the log in screen appears, then the display is connected to the EC520.

Some of the problems that you might see include:

Problem	Description	Resolution
The display does not start.	There is no power to the display.	Check connectors between the display and the EC520. Use the electrical schematic to check the pin numbers from the battery.
Using my own display, and not the TD5x0, I cannot connect to the EC520.	The first time you run Earthworks, there is no Wi-Fi connection with the EC520.	Connect a laptop, or similar, to the EC520 long platform harness "Display, Eth1 + CAN2" connector. On the laptop, open the Web Interface by typing address 192.168.168.1 into a browser window, log in, and set up the Wi-Fi access point on the EC520 in the Web Interface's Configure > Wi-Fi Network.
On startup, I received a warning to upgrade the display operating system.	An important operating system update for the display is available.	Contact your dealer to schedule the update.

4.4 TD540 display install

This section describes how to install the TD540 display. There are 2 installation options:

- With the magnetic display mount (P/N 110262-00) and Ram ratchet ball-clamp, for cab window installations.
- With the post mount (P/N 597-1269) and Ram ratchet ball-clamp, for mounting on a vertical pole, such as a grab bar.

WARNING – If you have a pacemaker and handle the magnetic display mount, the magnet can affect the function of the pacemaker and cause physical harm. Remain at least 30 cm (1 foot) from the magnetic display mount if you have a pacemaker.

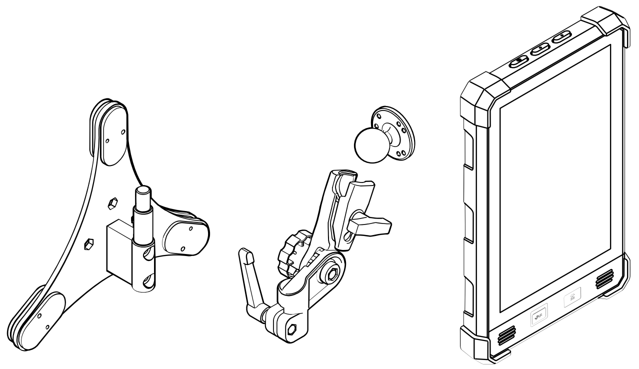
WARNING – The magnets on the magnetic display mount are strong. If parts of your hands get between two magnets when you move them, the magnets may suddenly move and pinch or clamp the skin. When the magnets are close together, position your hands so they are not between the two magnets.

NOTE – It is easier to perform this task with two people.

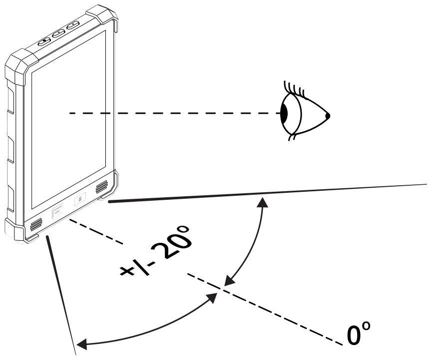
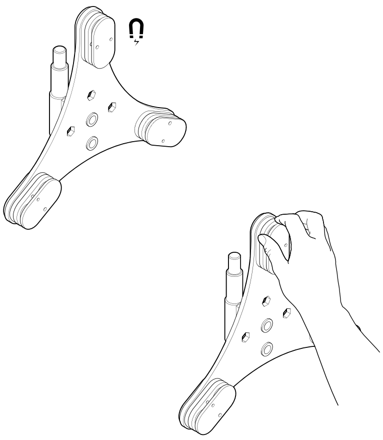
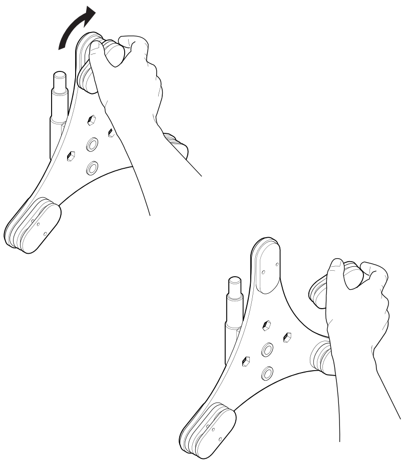
To select a route for the display cable, try various positions while considering:

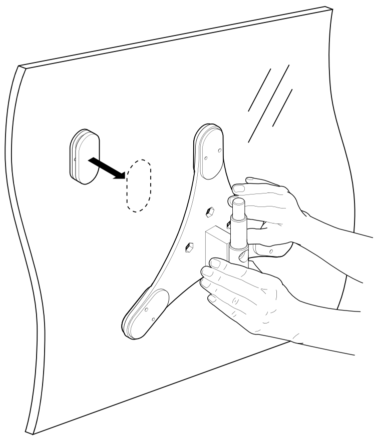
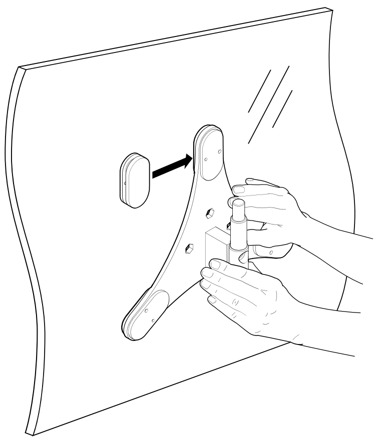
- Exposure to hazards, including:
  - mechanical damage by the operator
  - environmental hazards
  - heat
  - corrosive material
  - weather
- Existing grommeted entry points. When taking the cable outside the cab, use grommeted holes to minimize dust and moisture ingress to the cab.
- Maintaining free space between structures holding the display, or the display itself. For example, maintain free space between the display and windows that open and close, controls, and the view of the work area.
- Access to the AA510 audible alarm
- Avoid moving parts
- Avoid sharp edges
- Operator's preference for location and orientation

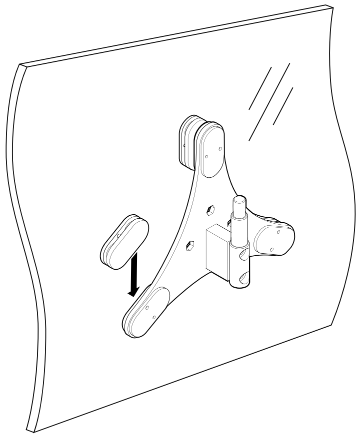
4.4.1 Option 1: Magnetic mount and display



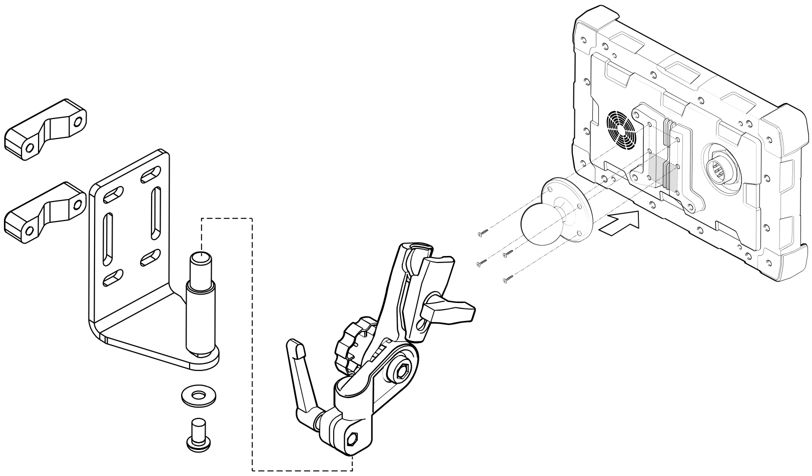
4.4.2 Attach magnetic mount and display

Steps	Description
	Decide on the location of the display.
	Hold the magnetic display mount and carefully twist the magnet and tri-arm in opposite directions.
	When the force between the magnets weakens, pull the tri-arm and magnet apart. Remove the padding and keep the magnets away from each other.

Steps	Description
Practice connecting and disconnecting.	Using two people and the padding, practice placing the tri-arm and magnet together. Keep the magnets close but not opposite each other with the padding in between, then rotate and slide the magnet until the magnets connect.
	On the machine, place the tri-arm magnetic assembly against the inside cab window in the required position. Check the post is aligned correctly and that the cables, Ram ratchet ball-clamp, and display are in the correct position. Remove the Ram ratchet ball-clamp and display before continuing. Hold the interior tri-arm firmly in the required position and make sure all three magnets touch the glass.
	On the outside of the glass, attach the first magnet. • Once the magnets connect they are very hard to disconnect. • Keep the magnet surfaces away from dirt or other metals. • Start with the outside magnet away from the inside magnet, then slide toward each other.

Steps	Description
	Repeat for all three magnets. Attach the Ram ratchet ball-clamp and then attach the display.

4.4.3 Option 2: Post mount and display



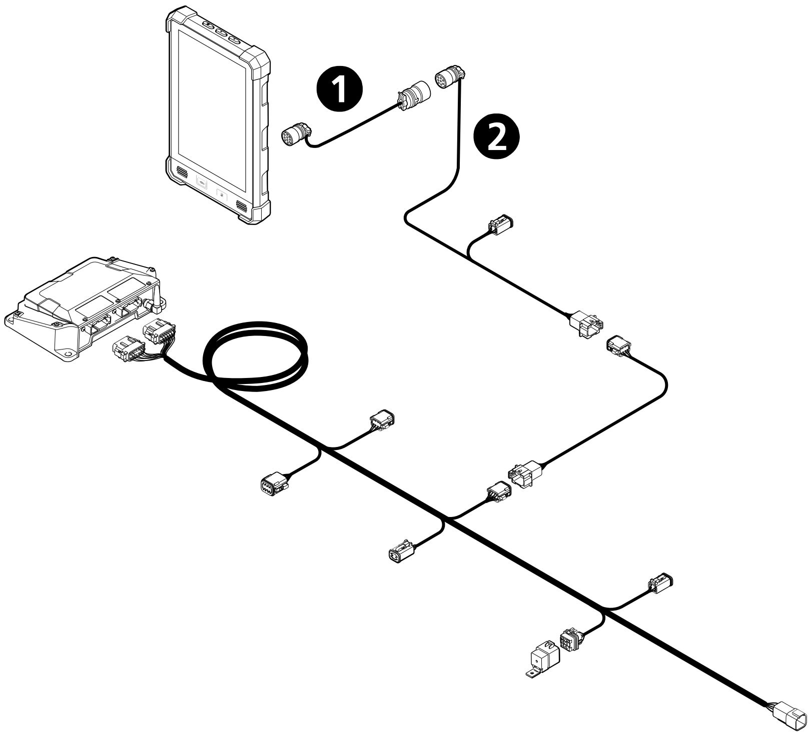
Attach the post mount

1. Connect the ball mount to the TD540.
2. Select a vertical post to mount the display to. The location should position the display within easy reach of the operator.
3. Use the appropriate M8 bolts and the V-blocks to attach the bracket to the post.
4. Fasten the RAM mount to the pole bracket.
5. Insert the ball mount into the RAM mount socket and fasten.

4.4.4 Connect the cables

1. Attach the TD540 adapter cable (P/N 150878-002, ❶) to the TD540 display. The adapter converts the display connector from 8-pin to 10-pin to match the display cable.

2. Attach the other end of the adapter cable to the TD5x0 display cable (P/N 150701-xxx, ❷).
3. Attach the display cable and any extension cables required to reach the EC520 long platform harness (P/N 150603-02).



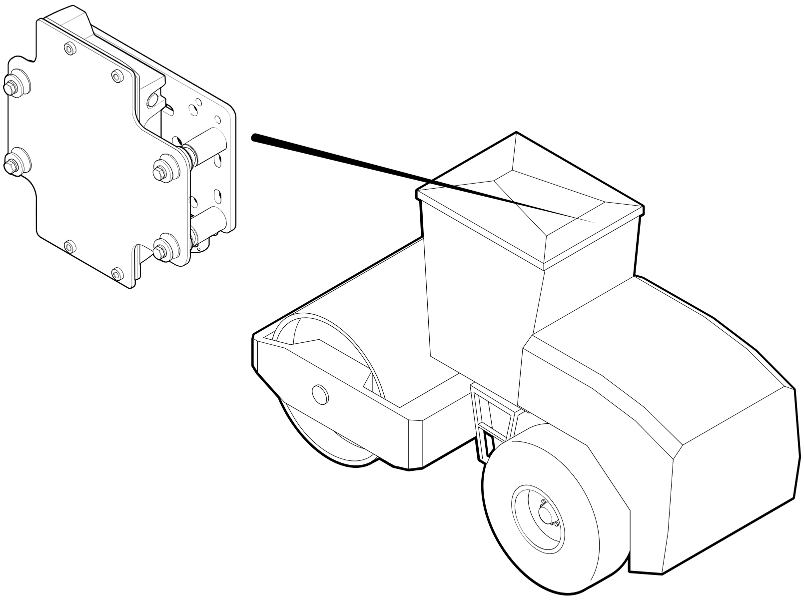
4.4.5 Validate

1. Turn on the ignition. The display starts automatically.
2. Tap the Earthworks icon. Earthworks opens.
3. If the log in screen appears, then the display is connected to the EC520.

Some of the problems that you might see include:

Problem	Description	Resolution
The display does not start.	There is no power to the display.	Check connectors between the display and the EC520. Use the electrical schematic to check the pin numbers from the battery.
On startup, I received a warning to upgrade the display operating system.	An important operating system update for the display is available.	Contact your system dealer to schedule the update.

4.5 SNM94x Connected Site Gateway install



> **NOTE –**Do not weld any components at this stage. Carefully mark the locations for the weld-mounted components, and then route the cables to make sure they will reach the components when they are installed.

> **NOTE –**Follow the connection diagram to check the cable reaches the proposed location of the bracket. Do not cut the cable. Use extension cables where needed.

The primary consideration is for operator and technician safety:

- Rotate the SNM94x Connected Site Gateway so that the connector is along the most convenient side and you can access the SIM card. If you rotate the bracket, move the stowage blank appropriately.

For optimum performance:

- Connect any antenna above the roof line with an uninterrupted view.
- The lengths of the cable between antenna and the SNM94x Connected Site Gateway are fixed. Do not use extension cables as extension cables reduce performance. Do not shorten the antenna cables as cutting cables violates some regulations.
- To prevent signal interference the antenna must be at least 200 mm away from any other transmitting antenna, 100 mm away from any receiving antenna (including SNR radio antennas), and higher than surrounding metal objects.
- Check that the radio cable reaches from the EC520 long platform harness to the radio bracket on the cab.
- Consider the location of the SNR rover radio connector and the route for the SNR radio cable.

To avoid damage to the SNM94x from shock and vibration, position the bracket on a solid part of the cab. Make sure the part is well braced as the extra weight of the radio and bracket can bend some machine parts.

For specific installation instructions, refer to the instructions provided with the radio bracket kit.

Validate

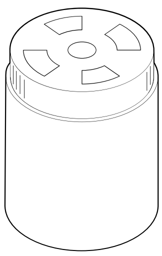
To check you have installed the SNM94x Connected Site Gateway and Wi-Fi antenna correctly, turn on the system.

- Complete the Setup section of the Install Assistant and enable the SNM94x.
- After a couple of minutes the SNM94x Connected Site Gateway will appear in the list of devices in the Web Interface's Monitor > Onboard Devices screen.
- Check that the Web Interface's Monitor > Onboard Devices screen shows that the Gateway Device is connected with a green tick.

4.6 AA510 audible alarm install

The AA510 is a required device that alerts the operator to events, for example avoidance zone warnings, whether they are on grade, above grade or below grade. The AA510 attaches to the system via a CAN bus.

4.6.1 Attach AA510



- Rubberized p-clamp P/N 97587-45
- Cable ties

To install:

1. Decide where to locate the AA510, P/N 503-0955.
2. Connect the AA510 to the EC520 long platform harness, as shown in the connection diagram.
3. Use P-clamp or cable ties to attach the AA510 to a fixed surface or component.

4.6.2 Validate

After connecting the display and running Earthworks, test the AA510 using the System Settings > Sound option in the Operator App.

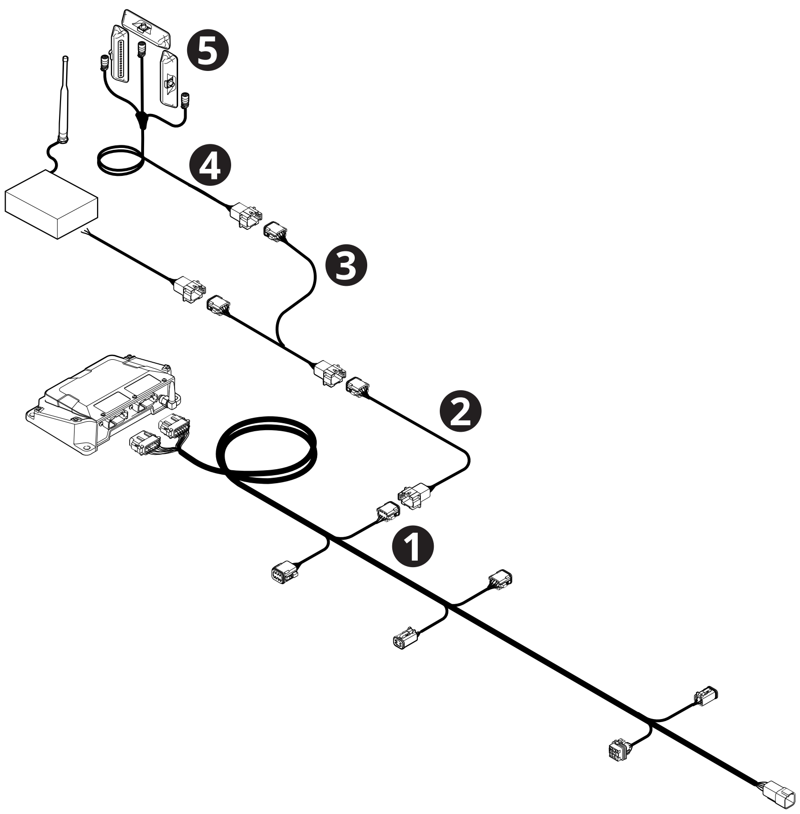
Some of the problems that you might see include:

Problem	Description	Resolution
AA510 too quiet or too loud.	The AA510 might seem loud in one machine and not loud enough in another machine because of where you locate the AA510 and the machine noise.	In the Operator App, open Sound Settings and edit the volume of the AA510 and the type of event that sets off the AA510.

4.7 External lightbars install (optional)

The optional external lightbars give extra guidance about being on grade, below grade, or above grade.

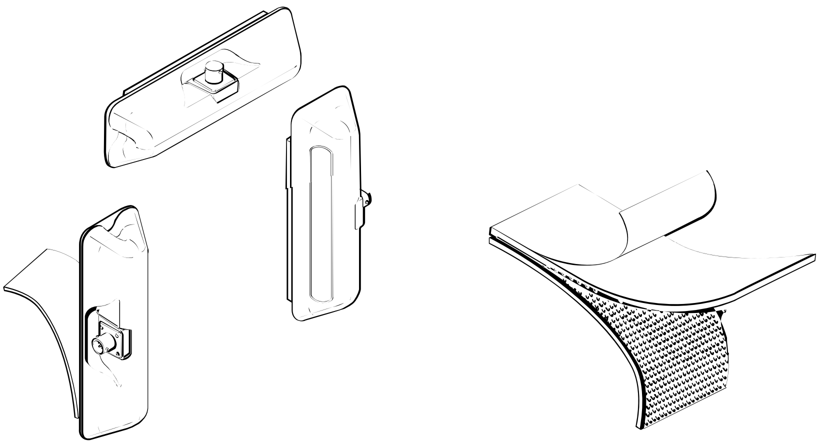
4.7.1 Connections, components, and cables



Part number	Item	Quantity	Description
150603-02	1	1	EC520 long platform harness
150418-xx	2	1	Extension cable

Part number	Item	Quantity	Description
150766-01	❸	1	8-pin splitter cable
54438-xx	❹	1	Lightbar cable
74706-01	❺	3	LB400 lightbars in lightbar kit
54990	–	2	Industrial strength hook and loop tape (Velcro), in the Cab/Platform kit

4.7.2 Locate and attach lightbars



Each lightbar is 174 mm (6.85 in) long, 53 mm (2.1 in) wide, and has a height of 32 mm (1.3 in).

To install:

1. Locate the Lightbar RS232 port on the EC520 long platform harness ❶. See 4.1.2 Cab components, cables and connections.
2. Connect an extension cable ❷ to the port.
3. Connect a splitter cable ❸ to the extension cable.
4. Connect the lightbar cable ❹ to the splitter cable.
5. Place the lightbars inside the machine cab using Velcro ❺. Ensure that the LEDs on the lightbars are visible to the operator.

If you remove the lightbars, place them in the foam-lined case for protection.

4.7.3 Validate

Do the following:

1. In the Web Interface, make sure you complete the entire Configure section, starting with the Install Assistant.

2. In Configure > External Lightbars, set the left, right, and center lightbars and select whether the lightbar mounting orientation is inverted.
3. In the Operator App, go to System Settings > Lightbars, use the slider to adjust the brightness of the lightbars.
4. Observe the lightbars during start up. Lightbars flash in a sequence when they start up.

The LEDs flash in the following sequence on start up:

LEDs on this lightbar	Flash
Left (drum tip cut/fill)	Bottom to top
Center (offline left/right)	Left to right
Right (drum tip cut/fill)	Top to bottom
All	Twice, simultaneously

Some of the problems that you might see include:

Problem	Description	Resolution
Lightbars do not illuminate	No power, or lightbar brightness is low.	Check that the display is on (to show power is on) and check the lightbar brightness is not set too low in the Operator App. Check the system harness between the EC520 and the lightbars.
Lightbars flash in the wrong sequence.	See sequences above.	The lightbars have been configured incorrectly. Open the Web Interface and check the configuration.

NOTE –For information about how lightbars work and how to operate them read the Operator's Guide, External Lightbars, available online or in the Earthworks Operator App.

4.8 Remote switch and CI5xx CAN interface install

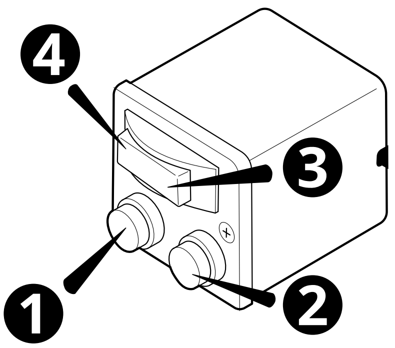
Both the CI510 and the CI520 CAN interfaces are supported. The CI510 supports up to 8 switches and the CI520 supports up to 4 switches.

The remote switch supports:

- Incrementing or decrementing the horizontal offset. Incrementing moves the line right relative to the direction of the line and decrementing moves the line left relative to the direction of the line.

- Incrementing or decrementing the Pass Shift count.

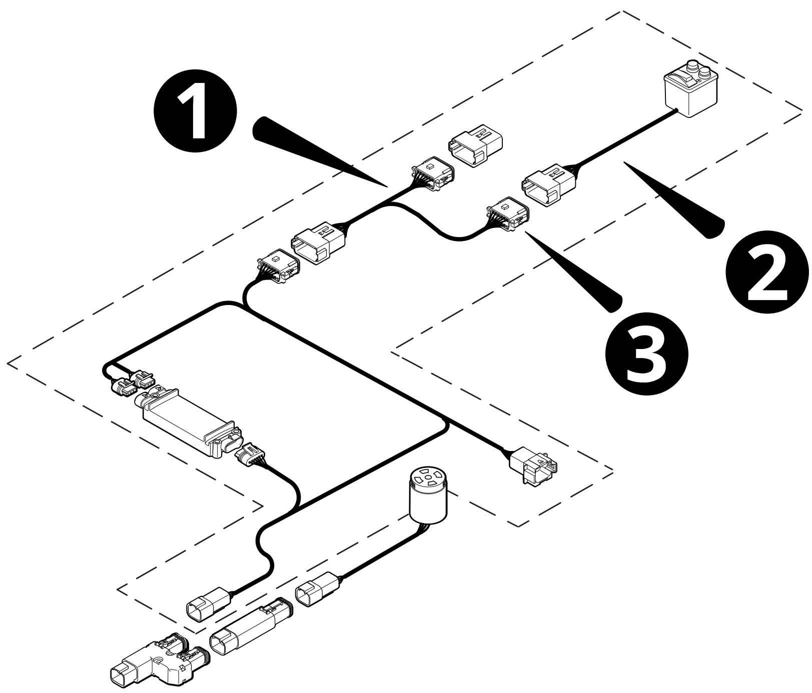
The functions on the remote switch are:



❶	Decrement Pass Shift count switch.	❷	Increment Pass Shift count switch.
❸	Increment Horizontal Distance D switch.	❹	Decrement Horizontal Distance D switch.

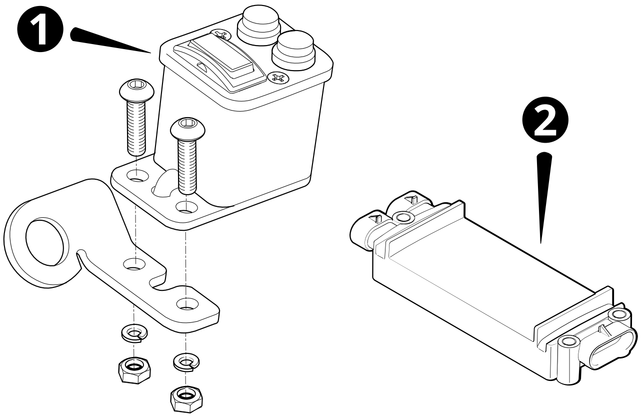
4.8.1 Remote switch installation with a CI510

Install the splitter cable ❶ (P/N 150864-01) between the switch harness (P/N 150754-01) and the switch. Ensure you connect the Pass Shift remote switch ❷ to Conn 2 ❸. Conn 1 is capped off:



CI510 interface module

You require a CI510 interface module to support the remote switch. It enables the remote switch to communicate with the system.



Item	Part number	Items	Description
1	150825-025	1	Remote switch assembly
2	109231-10	1	CI510 CAN interface module
	150754-01	1	Switch harness
	150864-01	1	Splitter cable

Install the CI510 switch harness

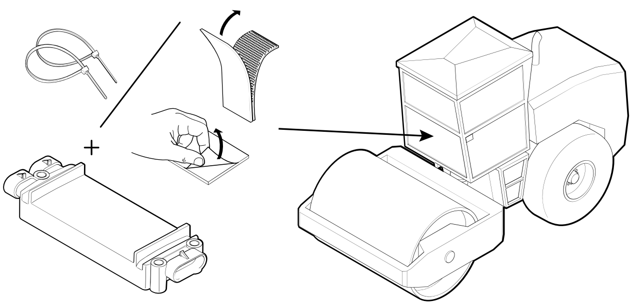
Select a route for the cable by considering:

- Access to the CI510

Install the CI510 CAN interface module

Have the following parts available:

- Dual locking strip
- Cable ties

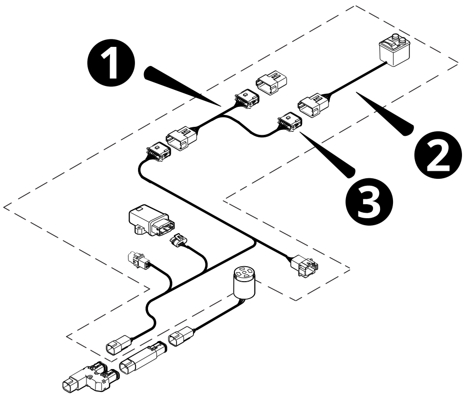


To install:

1. Connect the CI510 to the switch harness as shown in the connection diagram. See 4.1.2 Cab components, cables and connections.
2. Place the CI510 inside, or under, the cab. It is often behind the seat.
3. Use either zip ties or the Velcro supplied in the kit to secure the CI510, if required.

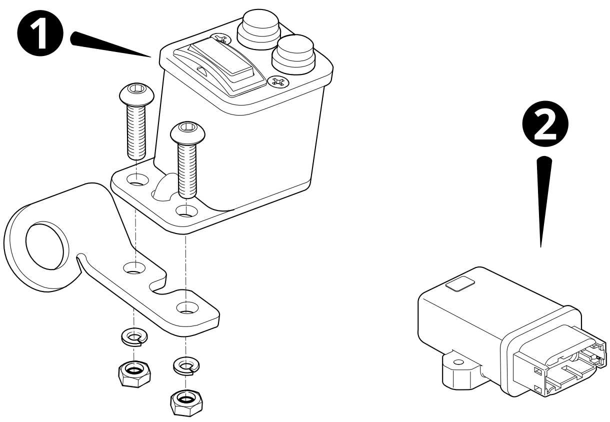
4.8.2 Remote switch installation with a CI520

Install the splitter cable ❶ (P/N 150911-01) between the switch harness (P/N 150820-01) and the switch. Ensure you connect the Pass Shift remote switch ❷ to Conn 2 ❸. Conn 1 is capped off:



CI520 interface module

You require a CI520 interface module to support the remote switch. It enables the remote switch to communicate with the system.



Item	Part number	Items	Description
❶	150825-025	1	Remote switch assembly
❷	132739-10	1	CI520 CAN interface module

Item	Part number	Items	Description
	150820-01	1	Switch harness
	150911-01	1	Splitter cable

Install the CI520 switch harness

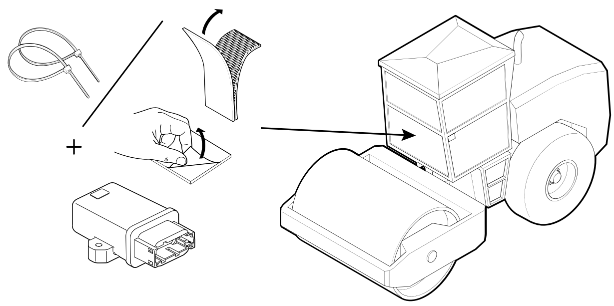
Select a route for the cable by considering:

- Access to the CI520

Install the CI520 CAN interface module

Have the following parts available:

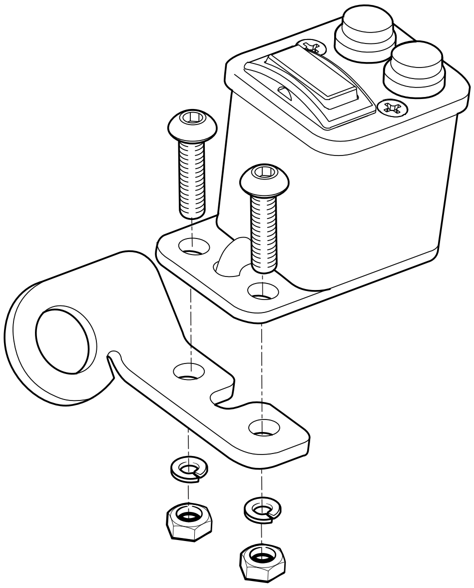
- Dual locking strip
- Cable ties



To install:

1. Connect the CI520 to the switch harness as shown in the connection diagram. See 4.1.2 Cab components, cables and connections.
2. Place the CI520 inside, or under, the cab. It is often behind the seat.
3. Use either zip ties or the Velcro supplied in the kit to secure the CI520, if required.

4.8.3 Install and connect the remote switch



To install:

1. Consider the operator's preference for location and orientation.

   > **NOTE –** If it is not practical to mount the remote switch on the control lever, you can make a custom mount.

2. Attach the supplied mounting tab to the base of the remote switch.

   > **NOTE –** If you are mounting to a joystick and your joystick cannot be detached from the control assembly, replace the mounting tab in the kit with the joystick handle mounting tab.

3. Attach the mounting tab to the operator's preferred location within the cab.
4. Route and secure the remote switch and connect the cable to the cab harness.

4.8.4 Validate the CI5xx installation

After connecting the display and running the Operator App, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI510 / CI520.

Some of the problems that you might see include:

Problem	Description	Resolution
Remote switch does not work.	The switch is not connected.	Check the switch connects to the system by opening the Web Interface and checking Monitor > Onboard Devices. Check the connections. Check that the cable from the remote switch is connected to the correct connector on the splitter cable between the switch and the CI5xx.

5

Compactor main harness install guide

In this guide:

5.1 Overview of the install

This guide describes the harness cables and connectors for a generic compactor system, and applies to a new machine or when replacing components on a machine currently using grade control.

5.2 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150473-020	1	Cable extension, 2m

5.3 Cable routing

One of the easiest ways to see how the system fits best on your machine is to drape the cables in possible locations, adding extension cables where needed and protecting any places where the cable will rub against edges.

Using a combination of splitters (P/N 150858-01), inline terminators (P/N 150857-01), and the cables listed in 5.2 Connections, components, and cables, connect the system hardware to the EC520 long platform harness.

> **NOTE –**Depending on the size of the machine, use more or fewer extension cables (P/N 150473-0x0, where x indicates the length of the cable in meters). The Generic Compactor Base Kit provides a 2 m extension cable.

During the install, ensure that:

- the cables do not interfere with, or catch on, any mechanical linkages. Protect cables as they pass any sharp metal edges.

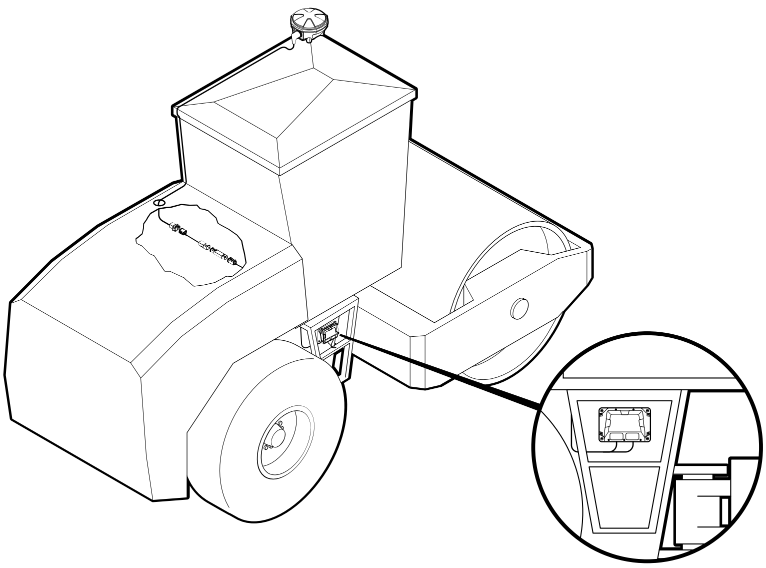
- you maintain access to any service maintenance panels.
- the cables are long enough to reach all of the system components.
- you avoid attaching the harness to the engine, engine components, or parts of the machine that get hot.
- extra cables are bundled and secured out of the way to prevent damage.
- cables are connected to something solid near the connector to limit stress on the connectors.

NOTE – Do not cut off any left over cabling, as this could affect system performance. Bundle, tie wrap and store any extra cabling in a convenient place on the machine.

5.3.1 Routing the main harness

- Install the main harness cable in a location where it is protected and has access to the cab and roof, for example, in the engine compartment of the machine under the cab. Coil excess cable and secure.
- Run the cable branch with the display connector into the cab.
- Run the cable branch with the radio and GNSS connector to a sheltered position toward the rear of the machine.

The following illustration shows a possible routing solution for a system with the GNSS receiver mounted on the front left of the cab roof:

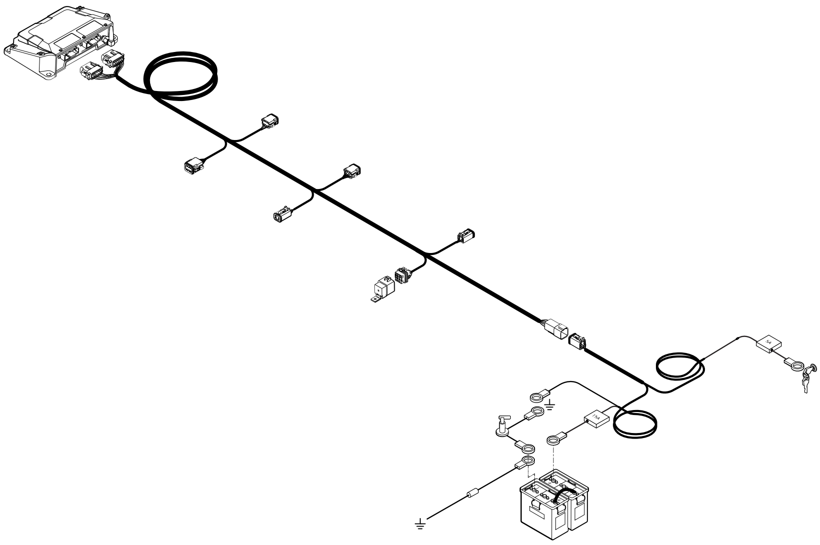


5.3.2 Deutsch connectors

The kit includes backshells and corrugated tubing for the EC520 Deutsch connectors. Use these to protect the connectors from water. When the backshell and tubing is installed, seal the end of the tubing with tape.

5.4 Power cable routing

NOTE – Turn off the master disconnect switch before connecting to the power source on the machine.



5.4.1 Power connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable
109333-12 *OR* 109333-24	1	Relay (12V) *OR* Relay (24V)
121022	1	Voltage protection diode*

* A voltage protection diode is only required if a CM310 compaction monitor is installed on a machine with a master disconnect switch.

TIP – While you are connecting the power, place a “Machine under service, do not turn on” tag on the battery isolation switch.

To connect:

1. Turn off the ignition and remove the battery isolation master key.
2. Route the power cable (P/N 150411-045) between the EC520 long platform harness and the battery.
3. Install the voltage protection diode, if required. See 6.2 Compaction sensor installation

4. Connect the power cable to the switched power supply.
5. Connect the relay, as shown in the connection diagram.
6. Verify that all wire connections are correct and tight.
7. Replace the battery isolation master key.

5.4.2 Connecting to machine power

NOTE –If you connect the system's ground (negative) connection to the negative terminal of the battery, and attempt to start the machine with the master disconnect switch open (or if the master disconnect switch or battery ground cable is defective), then the starter current will flow from a grounded component on the chassis to the battery terminal and damage the system. Make sure that you connect the system's ground connection to the machine frame.

TIP – While you are connecting the power, place a "Machine under service, do not turn on" tag on the battery isolation switch.

Constant power and key-switched power connections are made on the main run relay for the machine, using the power cable.

6

Compactor additional sensor install guide

In this guide:

6.1 Overview

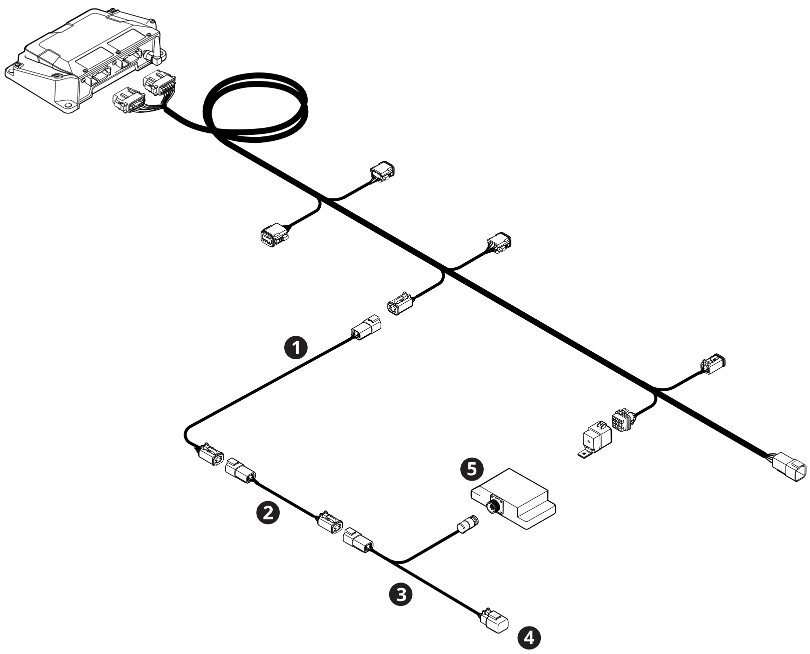
This guide describes how to install a compaction sensor on a compactor.

6.2 Compaction sensor installation

This section describes how to install the aftermarket CM310 compaction sensor.

NOTE –The compaction sensor may be pre-installed on Caterpillar machines.

6.2.1 Compaction sensor connections, components, and cables



Part number	Number	Quantity	Description
150473-xxx	❶	1	Extension cable
150839-005	❷	1	4-pin adapter cable
89773	❸	1	Compaction sensor cable
–	❹	1	Cap
64860-10	❺	1	Compaction sensor

6.2.2 Select a location for the compaction sensor bracket

Bolt or weld the bracket to the machine. Use existing holes where possible.

- Position the bracket on the hub of the machine drum.
- You must install the bracket on the:
  - inner side of the drum shock mounts
  - side of the drum that does not rotate
- Do not isolate the compaction sensor from the drum vibration.

6.2.3 Install the CM310 compaction sensor

1. Ensure that the sensor bracket is installed correctly.
2. Bolt the compaction sensor to the bracket.
3. Torque the bolts to 20 ± 2 N m (15 ± 1.5 lb-ft). Secure with Loctite.

Install the voltage protection diode

> **NOTE –**The voltage protection assembly is included in the CM310 compaction sensor kit, and is required only if the compaction sensor is installed and the machine has a master electrical disconnect switch.

1. Select a grounding point for the voltage protection diode cable (P/N 121022). Make sure that:
   - There is a guaranteed current path to the ground point of the battery. The best locations are the machine frame or a plate welded directly to the machine frame.
   - The cable reaches the grounded negative terminal of the battery. The grounded terminal is the terminal that connects to the master disconnect switch.
2. Connect the cable to the grounding point. Use the provided washer and nut to secure the diode to one of the following:
   - A hole in the machine frame. For thin metal, drill a 9.525 mm (0.375 in) hole; for thick metal, drill an 8.43 mm (0.332 in) hole and then tap the hole with a 3/8 – 24 tap.
   - The supplied square block (welded directly to the machine frame).

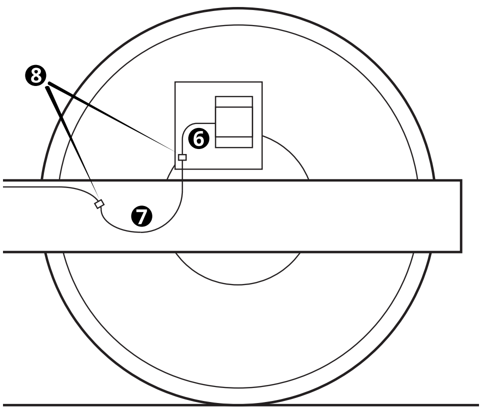
> **NOTE –**To ensure that the diode comes in complete contact with the metal surface, remove any existing paint. To keep the mounting surface from rusting and the diode from loosening, make sure that you apply rust prevention paint after you have secured the diode.

3. Torque the nut to:
   - 14 N m (125 lb-in) without lubricant
   - 12.5 N m (110 lb-in) lubricated
4. Use the provided nut to connect the lug end of the voltage protection diode cable to the grounded negative (–) terminal of the battery.
5. Verify that all wire connections are correct and tight.

6.2.4 Install the compaction sensor cables

> **NOTE –**Avoid creating tight bends in the compaction sensor cables. Where zip ties are used, avoid pulling them so tight that they compress the wires.

1. Connect an extension cable ❶ to BA-07 on the EC520 long platform harness.
2. Connect a 4-pin adapter cable ❷ to the extension cable.
3. Connect the compaction sensor harness ❸ to the adapter cable.
4. Place a cap on the unused connector on the harness ❹.
5. Connect the J1 connector on the harness to the sensor ❺.
6. Use a cable clamp ❽ to create a bend in the cable where it connects to the sensor, with a minimum bend radius of 60 mm (2.4 inches) ❻.
7. Use another cable clamp ❽ to create a bend with a radius of 170 mm (6.7 inches) ❼ to absorb the movement between the drum and the fork.
8. If the machine has existing hydraulic hose clamps, route the cable through them. Make sure not to create tight bends in the cable.



6.2.5 Validate

After connecting the display and running the Operator App, select System Status > Compaction Sensor. Ensure the sensor status is Active and the model is CM310.

7

Compactor GNSS install guide

In this guide:

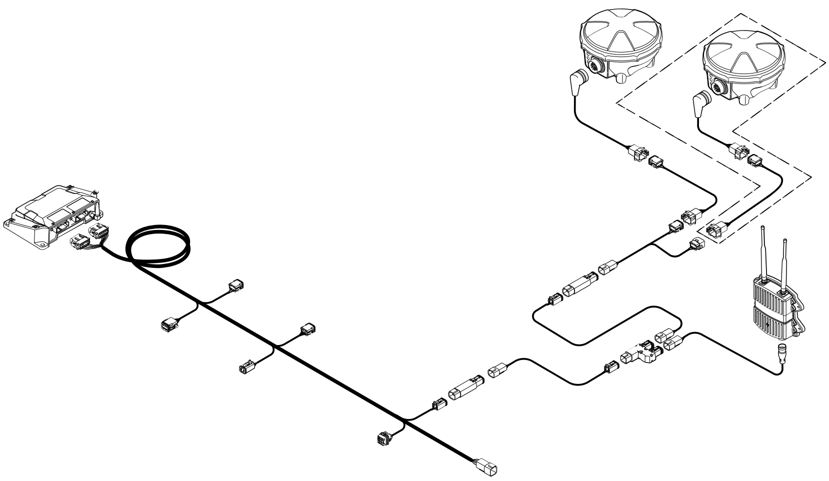
7.1 Overview of the install

This guide describes how to install either one or two GNSS receivers on the cab roof of a compactor.

Before you start to install the receivers:

- Position the machine on a flat and level surface.
- Review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and sensor installations.

7.2 GNSS connections, components, and cables

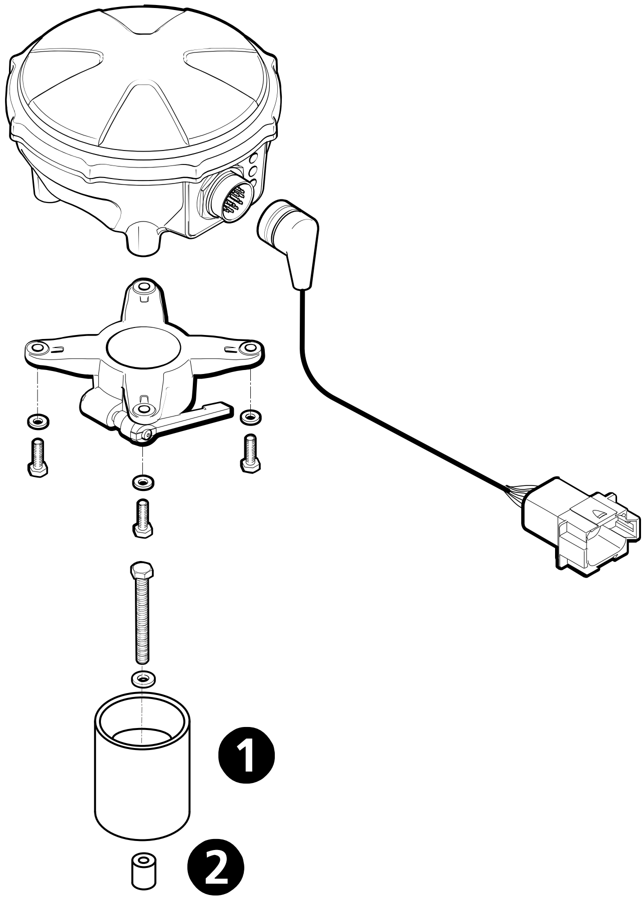


Part number	Quantity	Description
75698-00	1 or 2	Bracket – Roof Mount, with Hardware
–	1 or 2	MS992 or MS995 GNSS receiver

7.3 GNSS receiver install

The following guidelines describe how to mount a GNSS receiver on the cab roof. The roof mount kit (P/N 75698-00) includes a short mast (❶), and a boss (❷) for connecting the mast to the cab roof if necessary.

NOTE – Alternatively, you can use this mast with an MT9xx UTS target.



WARNING – Do not weld on or drill into any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

To install the receiver to the cab roof with an existing boss:

1. Locate an existing boss on the cab roof to mount the receiver to.
2. Use the supplied bolt to screw the mast to the boss.

3. Connect the quick release bracket to the receiver.
4. Mount the receiver on the mast and tighten the lever.
5. Connect the cable.

To install the receiver to a cab roof without an existing boss:

1. Weld the provided antenna pole boss (P/N 0794-1120) to a firm place on the cab roof.
2. Use the supplied bolt to screw the mast to the boss.
3. Connect the quick release bracket to the receiver.
4. Mount the receiver on the mast and tighten the lever.
5. Connect the cable.

7.4 Validate the GNSS system install

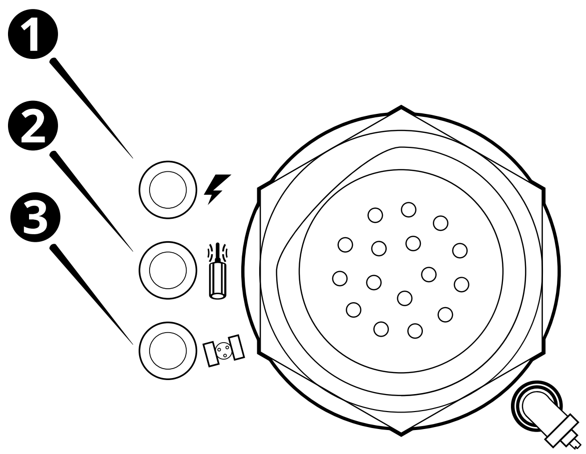
To validate the GNSS system installation:

- 7.4.1 Check GNSS receiver status indicators
- 7.4.2 Check radio power LED
- 7.4.3 Check device connections

7.4.1 Check GNSS receiver status indicators

The MS9xx GNSS receivers have three LEDs next to the harness connector. How these LEDs behave indicates the receiver status, as shown in the following figure.

In a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be solid.



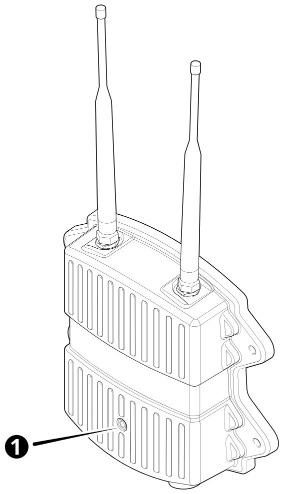
LED	Left	Right
❶ Power	Off – No power On – Power *	Off – No power On – Power *

LED	Left	Right
❷ Data link	Off – No corrections 1 Hz – Corrections †	Off – No corrections On – Corrections received from the left receiver †
❸ Satellite	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *

* required

† in a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be on (solid)

7.4.2 Check radio power LED



The radio is connected to power if the LED is permanently on or flashing.

7.4.3 Check device connections

In the Web Interface, navigate to Monitor > Onboard Devices, and check that the newly installed devices are shown as connected.

8

Compactor MT9xx UTS target install guide

In this guide:

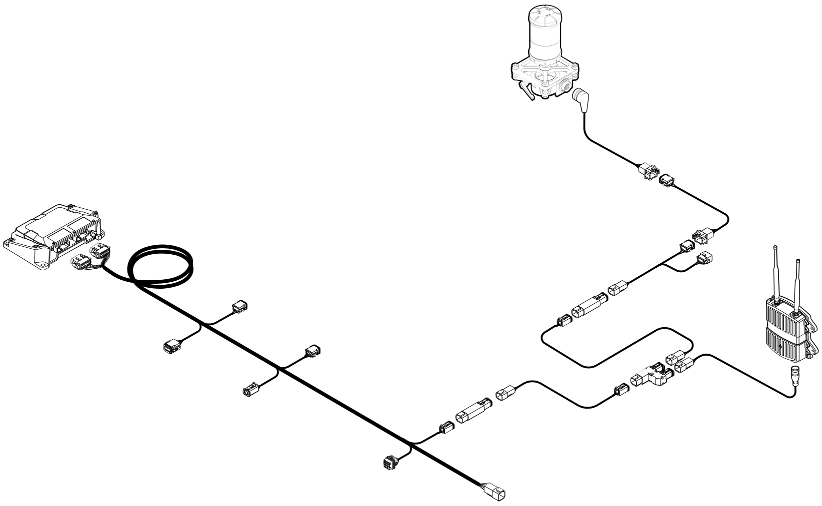
8.1 Overview of the install

This guide describes how to install an MT9xx UTS target on the cab roof of a compactor.

Before you start to install the target:

- Position the machine on a flat and level surface.
- Review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and sensor installations.

8.2 UTS connections, components, and cables

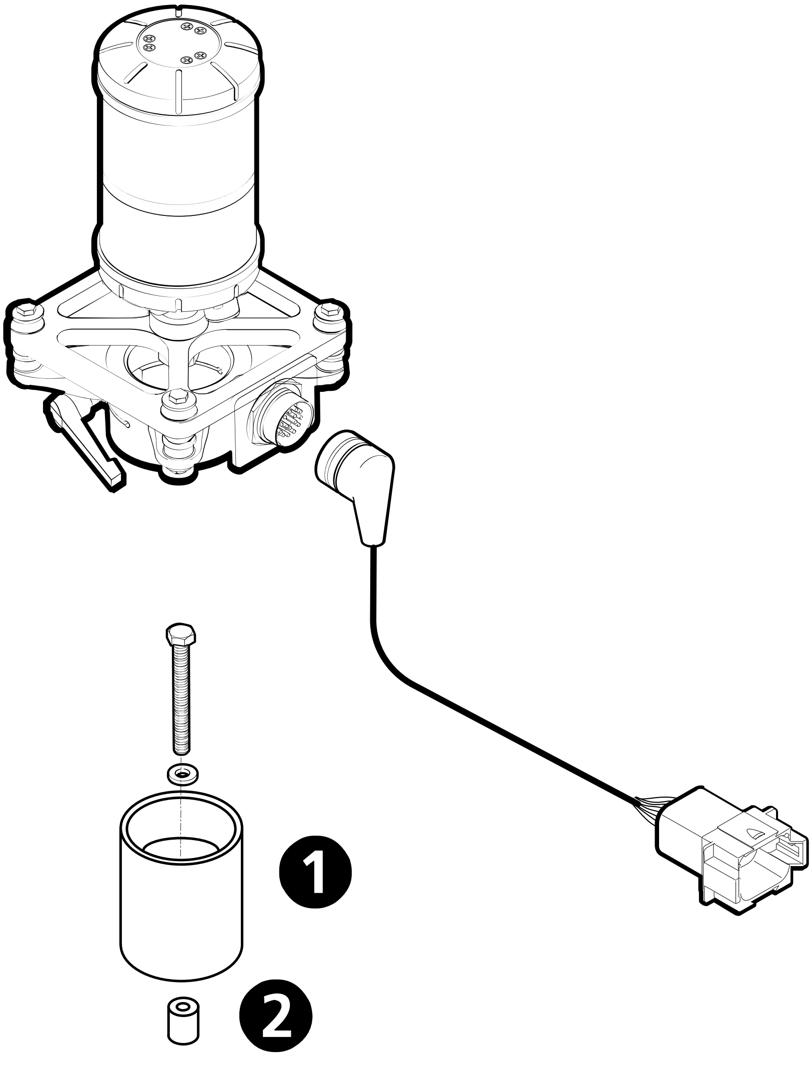


Part number	Quantity	Description
75698-00	1	Bracket – Roof Mount, with Hardware
59701-62	1	Kit - Machine Target, MT9xx w/ Shock Mount & Adaptor

8.3 UTS target install

The following guidelines describe how to mount a single MT9xx target on the cab roof.

The roof mount kit (P/N 75698-00) includes a short mast (❶), and a boss (❷) for connecting the mast to the cab roof if necessary. This mount can be used for GNSS receivers that use the quick release mounting bracket and the MT9xx target.



WARNING – Do not weld on or drill into any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

To install the target to the cab roof with an existing boss:

1. Locate an existing boss on the cab roof to mount the target to.
2. Use the supplied bolt to screw the mast to the boss.
3. Connect the quick release bracket to the target.
4. Mount the target on the mast and tighten the lever.
5. Connect the cable.

To install the target to a cab roof without an existing boss:

1. Weld the provided boss (P/N 0794-1120) to a firm place on the cab roof.
2. Use the supplied bolt to screw the mast to the boss.
3. Connect the quick release bracket to the target.
4. Mount the target on the mast and tighten the lever.
5. Connect the cable.

9

Compactor machine ECM install guide

In this guide:

9.1 Overview of the install

This guide describes how to connect the compactor harness to a machine ECM. This enables the system to receive additional data from the machine's sensors.

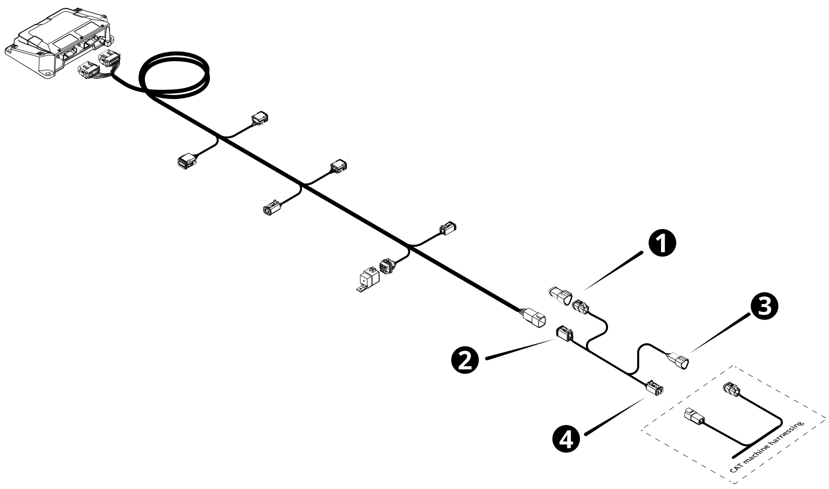
For more information, refer to the following document on Partners:

- Cable connection diagram: P/N 96608-05

Use the appropriate harness for your machine's interface:

- P/N 150867-01: Connects to both a 3-pin Deutsch connector and a 4-pin Deutsch connector on the machine

9.2 Installing Cat ECM cable P/N 150867-01



Part number	Quantity	Description
150867-01	1	Cat ECM cable

9.2.1 Connection process

Connect the Cat ECM cable to the system harness:

1. If necessary, disconnect the power harness from the system harness.
2. Remove the existing 3-pin CAN terminator from the Cat machine interface.
3. Connect the CAN terminator to the 3-pin Deutsch connector on the Cat ECM cable ❶.
4. Connect the 6-pin Deutsch connector on the Cat ECM cable ❷ to the system harness.
5. Connect the 3-pin Deutsch socket on the Cat ECM cable ❸ to the Cat machine interface.
6. Connect the 4-pin Deutsch socket on the Cat ECM cable ❹ to the Cat machine interface.

10

Torque charts

10.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (inches)										
Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1”
SAE 2	FT-LB	6	11	19	30	45	66	93	150	300
	Nm	8	15	26	41	61	89	126	203	407
SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
	Nm	12	24	42	68	102	149	203	339	786
SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
	Nm	18	39	62	102	156	224	305	502	1207
SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
	Nm	19	41	68	108	163	237	325	529	1302

10.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (mm)									
Grade	Torque	6	8	10	12	14	16	18	20
4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
	Nm	4	11	22	37	58	81	122	163
8.8	FT-LB	7	17	33	59	100	130	180	280
	Nm	10	23	45	80	136	176	244	380

			Nominal size (mm)							
	Grade	Torque	6	8	10	12	14	16	18	20
10.9	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
12.9	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–

Notices

See https://heavyindustry.trimble.com/earthworksreleases for the latest:

- Terms and conditions.
- Important safety information.
- Warranty and compliance information.

To provide feedback on this document, email feedback_documentation@trimble.com.